How AI Helped Build LinkLite

WedigTech Full Stack Developer assignment · Mini URL Shortener · Tool: Claude (Claude Code)

Summary

I defined the architecture and requirements up front. Claude wrote the code to that spec, ran the checks I asked for, and I directed the UI work. The full record is in AI_LOG.md.

Who did what

Area	My direction	What Claude did
Architecture	Chose Next.js + TypeScript + Tailwind, MongoDB/Mongoose only, a minimal folder structure, schema and API contract (201/400/500, 302, 404).	Scaffolded the project and wrote the model, connection helper, validation, short-code generator and route handlers.
Reliability	Required DB-enforced uniqueness with bounded retries, and secure random codes.	Implemented a unique index, a retry on duplicate-key errors, and Node `crypto.randomInt` codes.
Edge cases	Listed the cases to cover: empty, invalid, bad protocol, long URL, bad JSON, unknown code, restart, DB down.	Ran curl checks, 13 automated tests, lint, type-check and a production build; checked restart persistence and DB-down behavior.
Frontend	Asked for a responsive single page with loading/error states, copy/open and the last 5 links; later requested a modern UI redesign.	Built the form and result card, then a dark gradient redesign. Checked at desktop and 375px mobile.

Where AI got it wrong

- Claude exported `runtime/dynamic` from the route files, which this Next.js version rejects (HTTP 500). It spotted this in the dev server log and removed them.
- The generated `.gitignore` would have ignored `.env.example`; this was caught and fixed.
- Claude accidentally committed a log file once; it was removed and logs are now ignored.

Verified results

- Unit tests 9/9 and API tests 4/4 pass; lint, type-check and production build succeed.
- A short link still returns HTTP 302 with the correct Location after restarting the server.
- With MongoDB stopped, the API returns a safe 500 and recovers when MongoDB returns.

Not verified

- A forced short-code collision (retry loop), MongoDB Atlas, and the Copy button's "Copied" feedback.